PROJECT REPORT

PREDICTIVE ANALYTICS ON CUSTOMER CHURN PREDICTION

Submitted by

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Project Area: Predictive Analytics

Executive Summary

This project demonstrates how predictive analytics can help a subscription-based business identify customers who may discontinue a service. A synthetic dataset of 1,000 customer records was created for academic demonstration. It contains tenure, monthly charges, contract type, support tickets, internet service, payment method and add-on services. A Decision Tree classifier was trained and evaluated on a held-out test set. The synthetic dataset contains 289 churned customers (28.9%) and 711 retained customers (71.1%).

The test-set results are illustrative rather than evidence of performance on a real company's customer base. The model achieved an accuracy of 67.2%, precision of 45.3%, recall of 66.7%, F1-score of 53.9% and ROC-AUC of 0.780. These values provide a baseline for discussion and should not be treated as operational guarantees.

1. Introduction

Customer churn refers to the discontinuation of a product or service relationship. In subscription businesses, churn can reduce recurring revenue and increase the cost of acquiring replacement customers. Predictive analytics uses historical patterns to estimate the likelihood of future outcomes and can help organizations prioritize customer-retention activity.

This project applies a supervised classification approach to estimate whether a customer is likely to churn. The model output can support customer segmentation and targeted service interventions, while the final decision remains with the business team.

Objectives

• Prepare a structured customer dataset for churn analysis.

• Apply a Decision Tree classification model to estimate churn.

• Evaluate the model using standard classification metrics.

• Interpret customer-level risk patterns and propose retention actions.

2. Problem Statement

A business with a large customer base may struggle to identify customers who are at risk of leaving before cancellation occurs. The project addresses the following question:

Can customer profile and service-usage information be used to classify customers by churn likelihood and guide timely retention efforts?

Scope and limitations

The analysis is limited to a synthetic, generated dataset created for this report. It is not a sample of actual customers and does not represent any named telecom operator. The project demonstrates a reproducible analytics workflow; it does not establish causal relationships or validate a live business intervention.

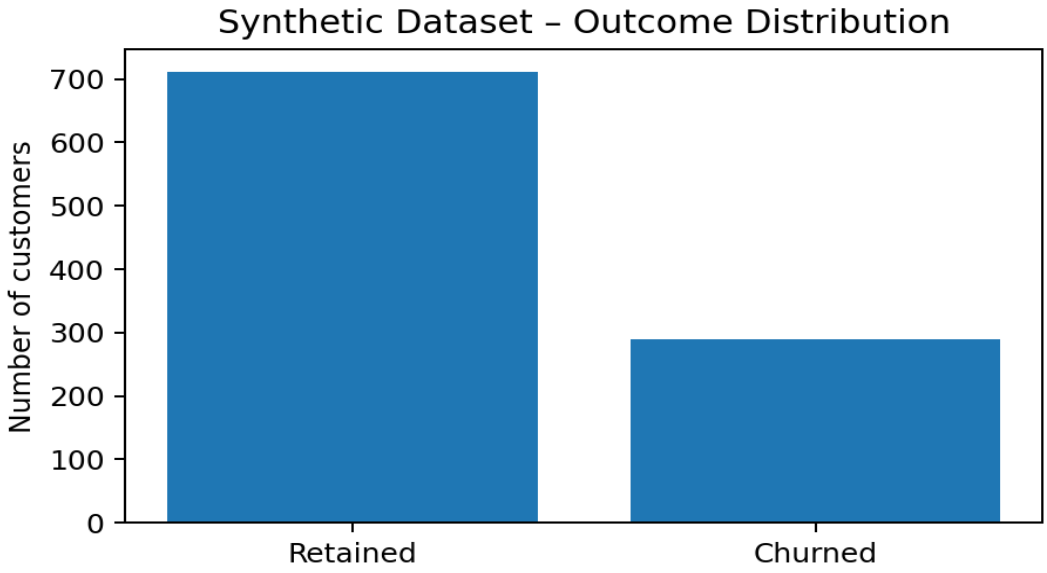
3. Dataset Description

The dataset contains 1,000 records and eight fields, including a unique customer identifier and the binary churn target. The CSV file is provided separately so the analysis can be reproduced.

Field	Description	Type
Customer_ID	Unique identifier for each customer	Identifier
Tenure_Months	Length of customer relationship	Numeric
Monthly_Charges_INR	Monthly service charge in INR	Numeric

Field	Description	Type
Contract_Type	Month-to-month, one year or two year	Categorical
Support_Tickets_6M	Support tickets raised in the previous six months	Numeric
Internet_Service	Fiber, DSL or no internet service	Categorical
Payment_Method	Customer payment channel	Categorical
Add_on_Services	Count of additional subscribed services	Numeric
Churn	1 = churned; 0 = retained	Target

Data preparation included checking the field types, retaining the customer ID for traceability but excluding it from model inputs, imputing missing values where needed, standardizing numeric inputs and one-hot encoding categorical variables. The generated dataset has no intentionally inserted missing values.



4. Methodology and Tools

Predictive method – Decision Tree

A Decision Tree classifier learns a sequence of feature-based splits that divide observations into groups with different target outcomes. It is relatively easy to explain because predictions can be traced through decision rules. The model used a maximum depth of five and a minimum leaf size of fifteen to limit unnecessary complexity.

Tools and technologies

• Python and pandas for dataset preparation.

• scikit-learn for preprocessing, train/test splitting, model training and evaluation.

• Matplotlib for visualizing dataset distribution and model metrics.

• CSV format for sharing the generated dataset.

5. Analytical Workflow

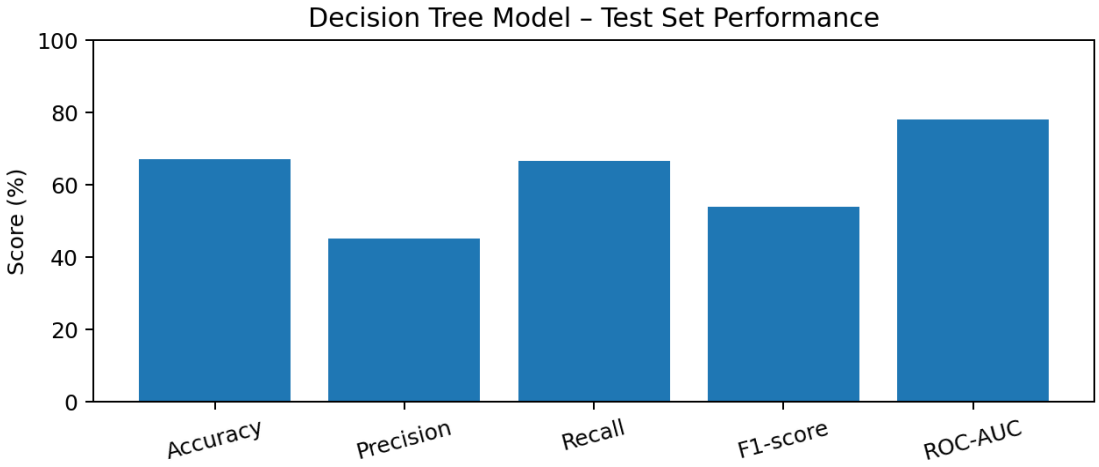
Step	Activity
1	Generate and inspect the synthetic customer dataset.

Step	Activity
2	Separate the churn target from explanatory variables.
3	Split the data into 75% training and 25% testing subsets using stratification.
4	Impute, scale numeric fields and encode categorical fields within a preprocessing pipeline.
5	Train a Decision Tree classifier on the training subset.
6	Predict churn labels and probabilities for the test subset.
7	Calculate classification metrics and review the confusion matrix.
8	Translate the results into cautious retention recommendations.

6. Model Results and Evaluation

The dataset was divided into training and test subsets. The test set contained 250 customer records. The model performance is summarized below.

Metric	Test result	Interpretation
Accuracy	67.2%	Share of all test records classified correctly.
Precision	45.3%	Share of predicted churn cases that were actual churn cases.
Recall	66.7%	Share of actual churn cases identified by the model.
F1-score	53.9%	Harmonic mean of precision and recall.
ROC-AUC	0.780	Ranking ability across probability thresholds.



Confusion matrix (actual rows; predicted columns): true negatives = 120, false positives = 58, false negatives = 24, and true positives = 48. The recall value indicates that the model identified about two-thirds of churn cases in this synthetic test sample, while precision indicates that some flagged customers were not churners.

A threshold change may increase recall but can also increase false positives. The appropriate threshold depends on the relative cost of missed churn and unnecessary retention contact.

7. Findings and Observations

• The synthetic data includes a mix of contract types, service categories and payment methods, allowing the workflow to demonstrate both numeric and categorical predictors.

• The model provides a useful baseline, but its accuracy alone is not sufficient to judge business value.

• False negatives matter because customers who are likely to leave may not receive timely attention.

• False positives also matter because unnecessary discounts or outreach can increase retention costs.

• The test results are specific to the generated data and random split; they should not be generalized to real customers.

8. Recommendations

• Use churn scores as a prioritization aid, not as a definitive prediction of an individual customer's intentions.

• Review service complaints, support history and recent changes in usage before deciding on an intervention.

• Design retention actions around customer needs; avoid automatically offering discounts to every flagged account.

• Monitor onboarding and early-tenure experience, and test whether service improvements reduce churn.

• Track precision, recall, F1-score and ROC-AUC over time, and review the confusion matrix at the chosen decision threshold.

• Validate the approach on real, authorized customer data before operational use, with privacy safeguards and appropriate access controls.

9. Conclusion

This project demonstrates a complete predictive-analytics workflow for customer churn classification using a Decision Tree model. A synthetic dataset was prepared, divided into training and test subsets, processed through a reproducible pipeline and evaluated with standard classification metrics. The results illustrate how customer information can be converted into churn-risk signals that support retention planning.

The key limitation is that the dataset is simulated. The model's reported performance is therefore a classroom demonstration, not a claim about a real organization. A production application would require representative historical data, careful validation, monitoring and responsible use of customer information.

References

• scikit-learn documentation: DecisionTreeClassifier and model evaluation metrics.

• pandas documentation: data structures and CSV input/output.

• Matplotlib documentation: plotting and visualization.

• Synthetic customer dataset generated specifically for this academic project.

Appendix: Sample Records

Customer ID	Tenure	Monthly ■	Contract	Tickets	Churn
CU0001	25	105.52	Month-to-month	1	No
CU0002	47	26.16	Two year	2	No
CU0003	9	44.20	Month-to-month	0	No
CU0004	67	40.29	Two year	2	No
CU0005	32	42.19	One year	2	No
CU0006	24	113.15	Month-to-month	0	Yes

Customer ID	Tenure	Monthly ■	Contract	Tickets	Churn
CU0007	47	41.07	Two year	1	No
CU0008	28	84.92	Month-to-month	1	No

Note: These are generated example records. The accompanying CSV contains the complete synthetic dataset.